

Assessment Tool 2 — Real-World Problem Solving

Q1 Smart Waste Recycling Plant System (5 marks)

Problem Statement

A recycling plant processes 5,000 kg of waste every day.

Waste Categories: Plastic, Paper, Metal, Electronic Waste

Efficiency Levels:

- $\geq 90\%$ Recycled $\rightarrow$ Excellent
- 75–89% $\rightarrow$ Good
- 50–74% $\rightarrow$ Average
- Below 50% $\rightarrow$ Poor

(a) Develop a C program using arithmetic operators to calculate recycled waste percentage and remaining waste. Use decision-making statements to determine plant efficiency.

(b) Use looping constructs to process waste data for 30 days. Use continue to skip maintenance days and break if the processing machine fails.

Code

```
#include <stdio.h>

int main() {
    float plastic, paper, metal, eWaste;
    float totalRecycled, percentage, remaining;

    printf("Enter recycled Plastic waste (kg): ");
    scanf("%f", &plastic);

    printf("Enter recycled Paper waste (kg): ");
    scanf("%f", &paper);

    printf("Enter recycled Metal waste (kg): ");
    scanf("%f", &metal);

    printf("Enter recycled Electronic waste (kg): ");
    scanf("%f", &eWaste);

    totalRecycled = plastic + paper + metal + eWaste;
```

```
percentage = (totalRecycled / 5000) * 100;
remaining = 5000 - totalRecycled;

printf("\nTotal Recycled Waste = %.2f kg", totalRecycled);
printf("\nRecycled Percentage = %.2f%%", percentage);
printf("\nRemaining Waste = %.2f kg", remaining);

if (percentage >= 90)
    printf("\nPlant Efficiency: Excellent");
else if (percentage >= 75)
    printf("\nPlant Efficiency: Good");
else if (percentage >= 50)
    printf("\nPlant Efficiency: Average");
else
    printf("\nPlant Efficiency: Poor");

return 0;
}
```

Sample Input

```
Enter recycled Plastic waste (kg): 1200
Enter recycled Paper waste (kg): 1300
Enter recycled Metal waste (kg): 1000
Enter recycled Electronic waste (kg): 1000
```

Sample Output

```
Total Recycled Waste = 4500.00 kg
Recycled Percentage = 90.00%
Remaining Waste = 500.00 kg
Plant Efficiency: Excellent
```

Q2 IT Company Project Performance Evaluation (2025-2026)

(5 marks)

Problem Statement

An IT company completed 20 software projects during the financial year 2025-2026. Each project is evaluated based on Quality Score (100 marks) and Delivery Delay (days).

Performance Criteria:

- Quality ≥ 90 and Delay ≤ 2 days $\rightarrow$ Excellent
- Quality 75-89 and Delay ≤ 5 days $\rightarrow$ Good
- Quality 60-74 $\rightarrow$ Average
- Below 60 $\rightarrow$ Needs Improvement

A) Develop a C program using arithmetic operators to calculate the average quality score and delivery percentage. Use decision-making statements to classify each project.

B) Use looping constructs to process all 20 projects. Use continue to ignore cancelled projects and break if management stops the evaluation.

Code

```
#include <stdio.h>
int main() {
    int i, quality, delay;
    int totalQuality = 0, delivered = 0;
    float averageQuality, deliveryPercentage;
    char cancelled, stop;

    for (i = 1; i <= 20; i++) {
        printf("\nProject %d\n", i);

        printf("Is project cancelled? (Y/N): ");
        scanf(" %c", &cancelled);

        if (cancelled == 'Y' || cancelled == 'y') {
            continue;
        }

        printf("Enter Quality Score: ");
        scanf("%d", &quality);

        printf("Enter Delivery Delay (days): ");
        scanf("%d", &delay);
```

```

    totalQuality += quality;

    if (delay <= 5)
        delivered++;

    if (quality >= 90 && delay <= 2)
        printf("Performance: Excellent\n");
    else if (quality >= 75 && quality <= 89 && delay <= 5)
        printf("Performance: Good\n");
    else if (quality >= 60 && quality <= 74)
        printf("Performance: Average\n");
    else
        printf("Performance: Needs Improvement\n");

    printf("Management stopped evaluation? (Y/N): ");
    scanf(" %c", &stop);

    if (stop == 'Y' || stop == 'y')
        break;
}

if (i > 1) {
    averageQuality = (float)totalQuality / (i - 1);
    deliveryPercentage = ((float)delivered / (i - 1)) * 100;

    printf("\nAverage Quality Score = %.2f", averageQuality);
    printf("\nDelivery Percentage = %.2f%%\n", deliveryPercentage);
}

return 0;
}

```

Sample Input

```

Project 1
Is project cancelled? (Y/N): N
Enter Quality Score: 95
Enter Delivery Delay (days): 2
Management stopped evaluation? (Y/N): N
... (repeats for Projects 2-20, or until "Y" is entered to stop) ...

```

Sample Output

```

Project 1
Is project cancelled? (Y/N):
Enter Quality Score:

```

Enter Delivery Delay (days):

Performance: Average

Management stopped evaluation? (Y/N):

Project 2

...

Average Quality Score = <based on entered projects>

Delivery Percentage = <based on entered projects>%

Q3 Semiconductor Chip Manufacturing System (5 marks)

Problem Statement

A semiconductor company manufactures 10,000 chips per day.

Quality Standards:

- Defect Rate $< 1\%$ $\rightarrow$ Grade A
- $1-3\%$ $\rightarrow$ Grade B
- $3-5\%$ $\rightarrow$ Grade C
- Above 5% $\rightarrow$ Reject Batch

A) Develop a C program using arithmetic operators to calculate the defect percentage and accepted chips. Use decision-making statements to assign quality grades.

B) Use looping constructs to process production for 30 days. Use continue for maintenance days and break if a machine breakdown occurs.

Code

```
#include <stdio.h>
int main() {
    int chips, defective, accepted;
    float defectRate;

    printf("Enter total chips produced: ");
    scanf("%d", &chips);

    printf("Enter defective chips: ");
    scanf("%d", &defective);

    defectRate = ((float)defective / chips) * 100;
    accepted = chips - defective;

    printf("\nDefect Percentage = %.2f%%", defectRate);
    printf("\nAccepted Chips = %d", accepted);

    if (defectRate < 1)
        printf("\nQuality Grade: Grade A");
    else if (defectRate <= 3)
        printf("\nQuality Grade: Grade B");
    else if (defectRate <= 5)
        printf("\nQuality Grade: Grade C");
    else
        printf("\nQuality Grade: Reject Batch");
}
```

```
return 0;  
}
```

Sample Input

Enter total chips produced: 10000

Enter defective chips: 250

Sample Output

Defect Percentage = 2.50%

Accepted Chips = 9750

Quality Grade: Grade B

Q4 Cloud Storage Subscription Management (2026) (5 marks)

Problem Statement

A cloud company offers storage plans:

- Basic – 100 GB
- Standard – 500 GB
- Enterprise – 2 TB

Extra storage costs ₹4 per GB. If storage usage exceeds 95%, a warning should be generated.

A) Develop a C program using arithmetic operators to calculate used storage, remaining storage, and extra charges. Use decision-making statements to determine storage status.

B) Use looping constructs to process multiple customers. Use continue for inactive accounts and break if server storage becomes full.

Code

```
#include <stdio.h>
int main() {
    int plan, used;
    int storage, remaining, extra;
    float charge;

    printf("Enter plan (1-Basic, 2-Standard, 3-Enterprise): ");
    scanf("%d", &plan);

    printf("Enter used storage (GB): ");
    scanf("%d", &used);

    if (plan == 1)
        storage = 100;
    else if (plan == 2)
        storage = 500;
    else
        storage = 2000;

    remaining = storage - used;

    if (used > storage) {
        extra = used - storage;
        charge = extra * 4;
    } else {
        extra = 0;
```

```
        charge = 0;
    }

    printf("\nTotal Storage = %d GB", storage);
    printf("\nUsed Storage = %d GB", used);
    printf("\nRemaining Storage = %d GB", remaining);
    printf("\nExtra Storage = %d GB", extra);
    printf("\nExtra Charge = Rs. %.2f", charge);

    if (used > storage * 0.95)
        printf("\nWarning: Storage usage exceeds 95%!");
    else
        printf("\nStorage Status: Normal");

    return 0;
}
```

Sample Input

Enter plan (1-Basic, 2-Standard, 3-Enterprise): 2
Enter used storage (GB): 480

Sample Output

Total Storage = 500 GB
Used Storage = 480 GB
Remaining Storage = 20 GB
Extra Storage = 0 GB
Extra Charge = Rs. 0.00
Warning: Storage usage exceeds 95%!